

Appendix CM-v19: Core Micro-Theorem Pack

Canonical v19 Pack: Metric Repair, Ext Invariance, Type IV Defect Calculus, and Audit Closure

Version CM-v19.0.0-r2

Atsushi Kobayashi

June 2026

Normative status. This appendix is subordinate to AK-HPDST v19.0.0 Main Chapters 1–8, Foundation Appendices A–G, and CR-v19.0.0-r2. It extracts small reusable theorems at exactly their proved strength. It does not enlarge the Core, create an external-domain theorem, repair a missing artifact numerically, or promote a specification, operational rule, Search artifact, or companion template to theorem evidence.

Central repair. The legacy CM assertions that hard positive-threshold deletion is globally non-expansive are rejected. Their canonical replacement is the sharp two-sided threshold-gap-safe theorem. The Type IV cokernel diagnostic is written with v , never normative Latin u . Revision r2 adds exact-invariance, terminal-generic defect calculus, and the pointwise finite-dimensional single-morphism interleaving theorem at their proved strengths.

Contents

Appendix CM-v19: Core Micro-Theorem Pack	1
1 Scope, dependency baseline, and claim discipline	3
2 Barcode algebra and categorical guard-rails	3
3 Window restriction, crop order, and repaired evaluation	4
4 Metric repair and comparison-mode calculus	5
5 Representative and Ext-edge micro-theorems	7
6 Terminal-generic Type IV micro-theorems	9
7 Restart, obligation, and countable-audit micro-theorems	10
8 Bridge-ready internal theorem	11
9 Exact-invariance and terminal-generic defect calculus	12
10 Single-morphism metric bridge	14
11 CM register closure and legacy migration	15
11.1 Canonical CM register summary	16
11.2 Revision and legacy migration rule	17

12 Explicit exclusions	17
13 Primary literature	18
14 Provenance and status	18

1 Scope, dependency baseline, and claim discipline

Policy 1.1 (CM-v19-001: Core micro-theorem boundary and no external upgrade). A CM-v19 item may be consumed only with its CM UID, declared role, evidence status, comparison policy, complete hypotheses, source Claim UIDs, artifact identities, and failure routes. A CM-v19 theorem is internal to the declared v19 Core regime and yields no external-domain conclusion without a separately proved and registered Interface or Bridge Theorem.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-001
Role / status: operational_policy / operational
Source claims: CR-G-001
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Definition 1.2 (Standing repaired Core regime). Fix a field k . Work in the constructible one-parameter regime $\text{Pers}_k^{\text{cons}}$. For a finite-type filtered object F , monitored degree i , right-open window W , and threshold $\tau > 0$, define

$$B_{i,W}(F) := W_W P_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

The standard bottleneck metric with the v19 finite/infinite-bar convention is used whenever metric comparison is invoked.

Definition 1.3 (Typed CM status). Every CM item records a document role, evidence status, invocation role, conformance status, claim-level comparison policy, admissible runtime modes, source binding, and structured failure routes. Missing evidence gives undefined; it is never silently encoded as a numbered obstruction, zero, pass, not_invoked, or not_compared.

2 Barcode algebra and categorical guard-rails

Lemma 2.1 (CM-v19-002: Infinite-bar threshold invariance). *For every finite $\tau > 0$,*

$$T_\tau^{\text{bar}}([a, \infty)) = [a, \infty).$$

An infinite bar never lies in a finite threshold band $[\tau - \varepsilon, \tau + \varepsilon]$.

Proof. Its threshold length is $+\infty$, while hard deletion removes only finite bars of length at most τ . □

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-002
Role / status: core_lemma / proved
Source claims: CR-A-002
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 2.2 (CM-v19-003: Hard threshold deletion idempotence). *For every barcode profile B and $\tau > 0$,*

$$T_\tau^{\text{bar}} T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(B).$$

Proof. After the first deletion no finite bar of length at most τ remains. □

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-003
Role / status: core_micro_theorem / proved
Source claims: CR-A-005
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 2.3 (CM-v19-004: Hard threshold composition). *For $\tau, \sigma > 0$,*

$$T_{\sigma}^{\text{bar}} T_{\tau}^{\text{bar}}(B) = T_{\max\{\tau, \sigma\}}^{\text{bar}}(B).$$

Proof. The composite deletes exactly the finite bars whose length is at most τ or at most σ , equivalently at most $\max\{\tau, \sigma\}$. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-004
Role / status: core_micro_theorem / proved
Source claims: CR-A-006
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 2.4 (CM-v19-005: Retained-profile threshold monotonicity). *If $0 < \tau \leq \sigma$, every bar retained at threshold σ is retained at threshold τ .*

Proof. A finite length greater than σ is greater than τ ; infinite bars survive both thresholds. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-005
Role / status: core_micro_theorem / proved
Source claims: CR-A-005
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 2.5 (CM-v19-006: Positive-threshold short bars are not Serre). *For $\tau > 0$, the class of constructible persistence modules whose bars all have finite length at most τ is not extension-closed in general. Hence hard positive-threshold deletion is not globally licensed as a Serre localization or exact reflector.*

Proof. At $\tau = 1$, the exact sequence

$$0 \rightarrow k_{[1,2)} \rightarrow k_{[0,2)} \rightarrow k_{[0,1)} \rightarrow 0$$

has short subobject and quotient but a length-two middle interval. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-006
Role / status: core_micro_theorem / proved
Source claims: CR-A-003, CR-NC-008
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

3 Window restriction, crop order, and repaired evaluation

Theorem 3.1 (CM-v19-007: Nested window restriction identity). *If $W' \subseteq W$, then at barcode-profile level*

$$W_{W'} W_W(B) = W_{W'}(B).$$

In particular, W_W is idempotent.

Proof. For each bar I , $(I \cap W) \cap W' = I \cap W'$. Empty intersections and multiplicities agree. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-007
Role / status: core_micro_theorem / proved
Source claims: CM-localproof; CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

Policy 3.2 (CM-v19-008: Fixed window-first repaired evaluation). The canonical evaluation order is

$$F \mapsto P_i(F) \mapsto W_W P_i(F) \mapsto T_\tau^{\text{bar}}(W_W P_i(F)).$$

Threshold clearance is measured after windowing and before deletion. A crop-after-collapse expression is not substituted unless CM-v19-010 is discharged.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-008

Role / status: operational_policy / operational

Source claims: CR-CH2-005, CR-A-013

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

Theorem 3.3 (CM-v19-009: No unconditional crop-collapse commutation). *In general,*

$$W_W T_\tau^{\text{bar}}(B) \neq T_\tau^{\text{bar}} W_W(B).$$

Proof. Take $B = \{[0, 10)\}$, $W = [0, 1)$, and $\tau = 2$. Deletion first retains the long bar and cropping produces $[0, 1)$; cropping first produces a length-one bar which is deleted. The same phenomenon occurs for an infinite bar on a bounded window. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-009

Role / status: core_micro_theorem / proved

Source claims: CR-A-010, CR-NC-010

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 3.4 (CM-v19-010: Exact crop-collapse criterion). *Let (B, W) be τ -window-adapted: every globally retained bar, including every infinite bar, is either disjoint from W or remains retained after clipping. Then, and only then,*

$$W_W T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}} W_W(B)$$

as barcode multisets including multiplicity.

Proof. Globally deleted finite bars remain deleted after clipping because clipping does not increase length. A globally retained bar contributes equally in both orders precisely under the stated all-retained-bars condition. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-010

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-A-011

Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-T3, FR-T5

4 Metric repair and comparison-mode calculus

Theorem 4.1 (CM-v19-011: Global non-expansiveness of hard deletion fails). *There exist barcodes B, C with*

$$d_B(B, C) < d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C).$$

Proof. Fix $0 < \eta < \tau/4$ and let

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Then $d_B(B, C) = 2\eta$. After deletion only the bar of B remains, so the distance is $(\tau + \eta)/2 > 2\eta$. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-011
Role / status: metric_repair_theorem / proved
Source claims: CR-A-004, CR-NC-007
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 4.2 (CM-v19-012: Sharp two-sided threshold-gap-safe stability). *Let B, C be finite barcodes, possibly with infinite bars. If $d_B(B, C) \leq \varepsilon$ and no finite bar of either barcode has length in the closed band*

$$[\tau - \varepsilon, \tau + \varepsilon],$$

then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The half-width ε is sharp for this two-sided certificate.

Proof. Choose an ε -matching. Infinite bars match only infinite bars and survive. A retained finite bar has length greater than $\tau + \varepsilon$. If its finite match were deleted, the two-sided closed-band hypothesis would force the matched length below $\tau - \varepsilon$, contradicting the fact that matched finite-bar lengths differ by at most 2ε . The argument is symmetric. Removing pairs deleted on both sides restricts the original matching to the retained barcodes with no larger cost. Sharpness follows from centered bars of lengths $\tau + c + \eta$ and $\tau - c - \eta$ for any $0 < c < \varepsilon$ and sufficiently small $\eta > 0$. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-012
Role / status: metric_repair_theorem / proved
Source claims: CR-A-008
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 4.3 (CM-v19-013: Scalarized-ledger-safe threshold classification). *Let $\mathfrak{d}_{\text{met}}$ lie in the declared metric-ledger quantale, let $\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}})$, and suppose the record proves*

$$d_B(B, C) \leq \varepsilon_B \quad \text{and} \quad \varepsilon_B < \text{gap}_\tau(B, C).$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B.$$

Proof. The strict inequality gives two-sided avoidance of the closed threshold band of half-width ε_B . Apply CM-v19-012. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-013
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-A-009
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5

Proposition 4.4 (CM-v19-014: Exact comparison is not a positive-radius robustness certificate). *An exact equality or profile isomorphism between two specified profiles does not imply a positive threshold-gap radius.*

Proof. A profile containing a finite bar of length exactly τ is exactly equal to itself, but has $\text{clear}_\tau = 0$. Thus exact comparison and positive-radius robustness are logically distinct. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-014
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-006
Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-T5

Policy 4.5 (CM-v19-015: Runtime comparison-mode trichotomy). Every consumed comparison resolves exactly one runtime mode:

exact, metric_gap_certified, not_compared.

The first requires an actual equality or profile isomorphism; the second requires a pre-threshold discrepancy bound and two-sided threshold-gap record; the third is admissible only when no comparison inference is consumed and cannot discharge an overlap, transport, coherence, representative, tower, or Restart obligation.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-015

Role / status: operational_policy / operational

Source claims: CR-CH1-002

Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared

Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 4.6 (CM-v19-016: Overlap transfer under a scalarized threshold-gap certificate). *For directly windowed pre-threshold overlap profiles $B_{i,W_{\alpha\beta}}^\alpha$ and $B_{i,W_{\alpha\beta}}^\beta$, suppose*

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

Then the repaired overlap profiles have bottleneck, and hence interleaving, distance at most $\varepsilon_{\alpha\beta,i}$ in the declared constructible regime.

Proof. Apply CM-v19-013 to the two directly windowed pre-threshold profiles and then the one-parameter isometry theorem. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-016

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-A-012, CR-E-001

Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified

Failure routes: FR-MISSING, FR-T3, FR-T5

5 Representative and Ext-edge micro-theorems

Theorem 5.1 (CM-v19-017: Windowed representative idempotence). *Let $C_{\tau,W}F$ be a verified window-normalized representative. If a second representative is constructed with the same window, monitored scope, and profile-equivalence policy, then*

$$C_{\tau,W}(C_{\tau,W}F) \simeq_{\text{PH}, I_{\text{mon}}} C_{\tau,W}F.$$

This is profile-level equivalence, not strict filtered equality or a global idempotent endofunctor.

Proof. Window normalization prevents further clipping, and CM-v19-003 makes threshold deletion idempotent on every monitored profile. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-017

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-B-007

Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 5.2 (CM-v19-018: Windowed representative threshold composition). *Let $\tau, \sigma > 0$. If all relevant representatives exist, are window-normalized, and use the same monitored scope and window, then*

$$C_{\sigma, W}(C_{\tau, W}F) \simeq_{\text{PH}, I_{\text{mon}}} C_{\max\{\tau, \sigma\}, W}F.$$

Proof. Use CM-v19-004 on each monitored profile. Window normalization prevents a second crop from creating new short bars. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-018
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-B-008
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 5.3 (CM-v19-019: Canonical two-term Ext^1 -edge identification). *For $A \in D^{[-1, 0]}(k\text{-mod})$, there is a canonical natural isomorphism*

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

Proof. Apply $\text{Hom}_{D^b}(-, k[1])$ to the canonical truncation triangle. Since k -vector spaces are semisimple, the outer Ext_k^1 and Ext_k^2 terms vanish, leaving the displayed natural isomorphism. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-019
Role / status: core_micro_theorem / proved
Source claims: CR-C-004
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5

Theorem 5.4 (CM-v19-020: Conditional one-way repaired PH_1 -to- Ext^1 bridge). *If $(F; W, \tau)$ has a complete eligible realization record and*

$$\text{Eval}_{1, W, \tau}(F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

Proof. Eligibility places $A = \mathcal{R}(C_{\tau, W}F)$ in $D^{[-1, 0]}(k\text{-mod})$ and supplies the edge-realization isomorphism between $H^{-1}(A)$ and the repaired degree-one residual edge. Vanishing of the repaired profile gives $H^{-1}(A) = 0$; CM-v19-019 then gives the Ext vanishing. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-020
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-C-008
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5

Theorem 5.5 (CM-v19-021: Reverse Ext^1 -to-repaired-profile implication fails). *Under the standing eligibility axioms alone,*

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0 \not\Rightarrow \text{Eval}_{1, W, \tau}(F) = 0.$$

Proof. Take a nonzero repaired interval profile and the constant-zero realization with constant-zero edge extractor. The amplitude and edge-realization requirements are satisfied at the zero extracted edge, and $\text{Ext}^1(0, k) = 0$, while the repaired profile remains nonzero. Reverse recovery therefore requires a separate faithfulness theorem. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-021
Role / status: core_micro_theorem / proved
Source claims: CR-C-012, CR-NC-006
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

6 Terminal-generic Type IV micro-theorems

Policy 6.1 (CM-v19-022: Actual artifact and terminal tameness are prerequisites for Type IV). Type IV is assigned exactly one status:

not_invoked, undefined, certified_zero, obstructed.

Numerical diagnostics are defined only after an actual tower comparison morphism, typed source and target, terminal-tame kernel and cokernel profiles, fixed $(W, \tau, I_{\text{mon}})$, and artifact provenance are present. Missing or ill-typed input yields undefined, never $(0, 0)$.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-022
Role / status: operational_policy / operational
Source claims: CR-CH5-003, CR-D-005, CR-NC-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 6.2 (CM-v19-023: Characterization of certified Type IV-free status). *A tower diagnostic record has status `certified_zero` if and only if its comparison artifact is valid, its kernel and cokernel defect profiles are terminal-tame, and*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

With the first two clauses valid, a nonzero pair gives `obstructed`.

Proof. These are exactly the validity, terminal-tameness, and numerical clauses defining the typed diagnostic record. □

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-023
Role / status: core_micro_theorem / proved
Source claims: CR-D-005
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 6.3 (CM-v19-024: Zero diagnostics plus no interior defects imply isomorphism). *Assume the defect profiles are terminal-tame and interval-decomposable. If the terminal-generic diagnostics vanish and the kernel and cokernel contain no interior or noncofinal interval summands, then the tower comparison artifact is an isomorphism in the declared profile category.*

Proof. The zero diagnostics exclude terminal-cofinal summands. The no-interior-defect hypothesis excludes all remaining interval summands. Thus kernel and cokernel vanish. □

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-024
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-006
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 6.4 (CM-v19-025: Full-isomorphism band implies terminal-generic stable band). *Every artifact-coherent threshold band on which each monitored tower comparison artifact is an isomorphism is a global terminal-generic Type IV stable band.*

Proof. Every isomorphism has zero kernel and cokernel, hence terminal-tame zero diagnostics and a certified_zero record at every threshold, assuming the remaining artifact fields are present. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-025
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-010
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

7 Restart, obligation, and countable-audit micro-theorems

Theorem 7.1 (CM-v19-026: Finite-horizon product-weighted Restart bound). *If transitions $0 \rightarrow 1, \dots, n-1 \rightarrow n$ satisfy*

$$r_{k+1} \geq \kappa_k r_k - \eta_k, \quad 0 < \kappa_k \leq 1,$$

then

$$r_n \geq \left(\prod_{j=0}^{n-1} \kappa_j \right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j \right) \eta_m.$$

Proof. Induct on n , substitute the previous bound into $r_{n+1} \geq \kappa_n r_n - \eta_n$, and distribute κ_n . $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-026
Role / status: core_micro_theorem / proved
Source claims: CR-E-005, CR-CH6-009
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 7.2 (CM-v19-027: Conditional countable Restart-summability principle). *Set $K_0 = 1$ and $K_n = \prod_{j=0}^{n-1} \kappa_j$. Suppose the initial reserve is positive, every transition is certified, every target record is complete with all invoked non-reserve clauses passing, no required record is undefined, and*

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Then every finite-stage reserve is positive and the ordered record family is an auditable sequential certificate chain. A uniform positive lower bound additionally requires $\inf_n K_n > 0$.

Proof. CM-v19-026 gives

$$\frac{r_n}{K_n} \geq r_0 - \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}} > 0.$$

The artifact and target-record hypotheses give reconstructibility. Uniform positivity does not follow if $K_n \rightarrow 0$. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-027
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-E-006, CR-CH6-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 7.3 (CM-v19-028: No scalar manufacture of non-scalar evidence). *No increase in a metric budget, threshold gap, reserve, or summability allowance can discharge a missing representative, actual morphism, eligibility record, terminal-tameness certificate, higher-coherence record, Claim UID, manifest field, or artifact identity.*

Proof. These obligations are typed as non-scalar atoms. The scalar ledger and the non-scalar obligation registry have disjoint semantic codomains; no scalar inequality constructs an absent typed witness. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-028
Role / status: core_micro_theorem / proved
Source claims: CR-F-001
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 7.4 (CM-v19-029: Non-compensation and dependency blocker propagation). *In the typed verdict algebra, a passing sibling cannot cancel a rejected or required undefined atom. If a required dependency is not passing, every dependent theorem-evidence obligation is blocked from pass until that dependency is repaired.*

Proof. Certificate readiness is a typed conjunction over dependency-closed obligations. Conjunction is downward closed: pass requires every required atom and dependency to pass. Therefore neither numerical surplus nor an unrelated passing atom can compensate for a blocker. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-029
Role / status: core_micro_theorem / proved
Source claims: CR-F-004, CR-F-006
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

Theorem 7.5 (CM-v19-030: Finite prefixes do not certify countable readiness). *A finite prefix of a countable Restart sequence, overlap atlas, registry, or manifest cannot certify the complete countable object unless a separate theorem reduces the infinite obligation to finite data.*

Proof. Any finite prefix omits later entries and their dependencies. Without a reduction theorem, omitted required obligations remain unchecked, so complete readiness is undefined rather than passed. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-030
Role / status: core_micro_theorem / proved
Source claims: CR-F-012, CR-GAPP-006
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

8 Bridge-ready internal theorem

Theorem 8.1 (CM-v19-031: Eventually constant repaired tower with apex agreement). *Let*

$$Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow \cdots$$

be a repaired profile tower in a category admitting the declared colimit. Suppose there exists N such that every transition after N is an isomorphism, the apex profile Q_∞ is equipped with a declared isomorphism to the stabilized profile Q_N , and the canonical colimit-to-apex artifact is the induced comparison. Then that artifact is an isomorphism. If the Type IV input is terminal-tame and valid, its diagnostics are $(0, 0)$.

Proof. The colimit of an eventually constant diagram is canonically isomorphic to its stabilized value. Composing with the declared apex agreement gives an isomorphism. Its kernel and cokernel vanish, so the terminal-generic diagnostics vanish. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-031
Role / status: conditional_core_micro_theorem / conditional
Source claims: CM-localproof;CRpromotionpending
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T4, FR-T5

Remark 8.2 (Bridge boundary). CM-v19-031 is an internal bridge-ready theorem. It does not prove that any arithmetic, PDE, geometric, categorical, or cryptographic tower realizes into such a diagram. That transport is a separate Interface or Bridge Theorem.

9 Exact-invariance and terminal-generic defect calculus

Theorem 9.1 (CM-v19-032: Representative-invariance of eligible Ext^1 discharge). *Let $C_{\tau,WF}$ and $C'_{\tau,WF}$ be admissible representatives with eligible realizations*

$$A := \mathcal{R}(C_{\tau,WF}), \quad A' := \mathcal{R}(C'_{\tau,WF})$$

in $D^{[-1,0]}(k\text{-mod})$. Suppose an isomorphism $\eta : A \xrightarrow{\sim} A'$ is compatible with the declared degree-one edge-realization records. Then precomposition induces an isomorphism

$$\eta^* : \text{Ext}^1(A', k) \xrightarrow{\sim} \text{Ext}^1(A, k),$$

and consequently

$$\text{Ext}^1(A, k) = 0 \iff \text{Ext}^1(A', k) = 0.$$

Thus an eligible categorical discharge is invariant under an artifact-backed change of realization-equivalent representative.

Proof. The contravariant functor $\text{Hom}_{D(k)}(-, k[1])$ sends isomorphisms to isomorphisms. Compatibility of the edge records ensures that the two Ext groups discharge the same declared repaired degree-one clause rather than merely two unrelated derived objects. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-032
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-007
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 9.2 (CM-v19-033: Exact-conjugacy invariance of Type IV diagnostics). *Let $\phi : M \rightarrow N$ and $\phi' : M' \rightarrow N'$ be actual comparison morphisms and suppose there are isomorphisms $\alpha : M \xrightarrow{\sim} M'$, $\beta : N \xrightarrow{\sim} N'$ such that*

$$\beta \circ \phi = \phi' \circ \alpha.$$

Then

$$\ker \phi \cong \ker \phi', \quad \text{coker } \phi \cong \text{coker } \phi'.$$

Whenever both records are terminal-tame on the same monitored scope,

$$\mu(\phi) = \mu(\phi'), \quad \nu(\phi) = \nu(\phi'),$$

and the corresponding total collapse diagnostics agree degreewise and after aggregation.

Proof. The isomorphism α restricts to the kernels, and β descends to the cokernels. Terminal-generic dimension is invariant under persistence-module isomorphism. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-033
Role / status: core_micro_theorem / proved
Source claims: CR-D-012
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 9.3 (CM-v19-034: Terminal-generic index identity). *Let $\phi : M \rightarrow N$ be an actual morphism between terminal-tame persistence modules with finite-dimensional stabilized tails. Then*

$$\mu(\phi) - \nu(\phi) = \text{tgdim}(M) - \text{tgdim}(N).$$

Proof. Choose a common terminal parameter after which M, N , and the defect profiles are constant up to the declared tail identifications. The induced finite-dimensional map $\phi_\infty : M_\infty \rightarrow N_\infty$ satisfies

$$\dim \ker \phi_\infty - \dim \text{coker } \phi_\infty = \dim M_\infty - \dim N_\infty$$

by rank-nullity. These four dimensions are precisely the declared terminal-generic dimensions. □

Corollary 9.4 (Balanced-tail consequence). *If $\text{tgdim}(M) = \text{tgdim}(N)$, then $\mu(\phi) = \nu(\phi)$. In particular, under the valid terminal-tame record, either one vanishes if and only if the other vanishes.*

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-034
Role / status: core_micro_theorem / proved
Source claims: CM-localproof; CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 9.5 (CM-v19-035: Direct-sum additivity of Type IV diagnostics). *For terminal-tame actual morphisms $\phi_1 : M_1 \rightarrow N_1$ and $\phi_2 : M_2 \rightarrow N_2$,*

$$\mu(\phi_1 \oplus \phi_2) = \mu(\phi_1) + \mu(\phi_2), \quad \nu(\phi_1 \oplus \phi_2) = \nu(\phi_1) + \nu(\phi_2).$$

The same statement holds for every finite direct sum and for the declared finite aggregation over monitored degrees.

Proof. Kernels and cokernels commute with finite biproducts in the persistence-module abelian category:

$$\ker(\phi_1 \oplus \phi_2) \cong \ker \phi_1 \oplus \ker \phi_2, \quad \text{coker}(\phi_1 \oplus \phi_2) \cong \text{coker } \phi_1 \oplus \text{coker } \phi_2.$$

Terminal-generic dimension is additive on finite direct sums. □

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-035
Role / status: core_micro_theorem / proved
Source claims: CM-localproof; CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 9.6 (CM-v19-036: Composition subadditivity of Type IV diagnostics). *Let*

$$M \xrightarrow{f} N \xrightarrow{g} P$$

be terminal-tame actual morphisms on a common terminal scope. Then

$$\mu(gf) \leq \mu(f) + \mu(g), \quad \nu(gf) \leq \nu(f) + \nu(g).$$

Consequently, two valid terminal-generically clean comparisons have a clean composite.

Proof. On a common stabilized tail these are finite-dimensional linear maps. The map $\ker(gf) \rightarrow \ker g$, $x \mapsto f(x)$, has kernel $\ker f$, giving the kernel inequality. Also

$$\text{rank}(gf) \geq \text{rank}(f) + \text{rank}(g) - \dim N,$$

which rearranges to the cokernel inequality. Tail dimensions equal the terminal-generic diagnostics. $\square$

Remark 9.7 (No converse). Cleanliness of gf does not force cleanliness of f or g ; defects may be killed or bypassed by composition.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-036
Role / status: core_micro_theorem / proved
Source claims: CM-localproof;CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 9.8 (CM-v19-037: Snake-lemma control over short exact comparison diagrams). *Consider a commutative diagram of persistence modules with exact rows*

$$\begin{array}{ccccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' & \rightarrow & 0 \\ & & \downarrow \phi' & & \downarrow \phi & & \downarrow \phi'' & & \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' & \rightarrow & 0. \end{array}$$

Assume the six modules and the vertical defect profiles are terminal-tame on a common terminal scope. Then

$$\mu(\phi) \leq \mu(\phi') + \mu(\phi''), \quad \nu(\phi) \leq \nu(\phi') + \nu(\phi'').$$

In particular, terminal-generic cleanliness of both outer vertical maps implies cleanliness of the middle map.

Proof. Pointwise exactness and the snake lemma give

$$0 \rightarrow \ker \phi' \rightarrow \ker \phi \rightarrow \ker \phi'' \rightarrow \text{coker } \phi' \rightarrow \text{coker } \phi \rightarrow \text{coker } \phi'' \rightarrow 0.$$

Pass to a common stabilized tail and take finite-dimensional vector-space dimensions. Exactness bounds the middle kernel and cokernel dimensions by the sums of their outer counterparts. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-037
Role / status: conditional_core_micro_theorem / conditional
Source claims: CM-localproof;CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

10 Single-morphism metric bridge

Definition 10.1 (Finite-shift triviality). For $\epsilon \geq 0$, a persistence module E is ϵ -trivial if its translation morphism $E \rightarrow E(\epsilon)$ is zero.

Theorem 10.2 (CM-v19-038: Single-morphism characterization of one-parameter interleaving). *Let M, N be pointwise finite-dimensional one-parameter persistence modules and let $\delta \geq 0$. Then M and N are δ -interleaved if and only if there exists a morphism*

$$f : M \longrightarrow N(\delta)$$

whose kernel and cokernel are both 2δ -trivial.

Proof. This is Corollary 6.6 of Bauer and Lesnick [1]. Its one-parameter and pointwise finite-dimensional hypotheses are part of the theorem and are not silently weakened. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-038
Role / status: core_micro_theorem / proved
Source binding: Bauer--Lesnick2015, Corollary6.6; CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5

Theorem 10.3 (CM-v19-039: Finite-trivial defect implies terminal-generic cleanliness). *Let $f : M \rightarrow N$ be an actual morphism whose kernel and cokernel are interval-decomposable, terminal-tame, and ϵ -trivial for some finite ϵ . Then*

$$\mu(f) = \nu(f) = 0.$$

Consequently, a morphism satisfying the defect hypothesis of CM-v19-038 is terminal-generically clean whenever it is used as the declared Type IV comparison artifact on the same scope.

Proof. A terminal-cofinal interval summand has nonzero structure maps across every finite shift sufficiently far out. Such a summand cannot occur in an ϵ -trivial module. Hence neither defect profile has a terminal-cofinal summand, so both terminal-generic dimensions vanish. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-039
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-004, CR-D-005; Bauer--Lesnick2015
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5

Theorem 10.4 (CM-v19-040: Terminal-generic cleanliness does not imply quantitative interleaving). *For every $\delta > 0$, terminal-generic zero of an actual comparison morphism does not imply that its kernel and cokernel are 2δ -trivial, and does not imply a δ -interleaving.*

Proof. Choose $L > 2\delta$, let $M = 0$, $N = k_{[0,L]}$, and take the zero morphism $f : M \rightarrow N(\delta)$. Its kernel is zero and its cokernel is a finite interval module, so both terminal-generic diagnostics vanish. However, the cokernel is not 2δ -trivial because its bar length is greater than 2δ . Moreover

$$d_I(M, N) = d_B(\emptyset, \{[0, L]\}) = L/2 > \delta.$$

Thus Type IV cleanliness is strictly weaker than a bounded-defect interleaving certificate. $\square$

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-040
Role / status: core_micro_theorem / proved
Source claims: CM-localcounterexample; CRpromotionpending
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5

11 CM register closure and legacy migration

Policy 11.1 (CM-v19-041: CM register closure and source-binding discipline). A CM snapshot is register-closed only when every consumed CM UID is unique, its source Claim UIDs or external primary-source locators resolve, every conditional hypothesis is discharged, the comparison policy is compatible with the runtime mode, failure routes are structured, and no deprecated or rejected legacy CM item is used as theorem evidence. An imported theorem is consumed only inside its stated source regime.

CM UID: ak-hpdst:19.0.0:CM-v19.0.0-r2:CM-v19-041
Role / status: operational_policy / operational
Source claims: CR-GAPP-002, CR-GAPP-003, CR-GAPP-018
Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

11.1 Canonical CM register summary

CM ID	Title	Status	Comparison policy
CM-v19-001	Core micro-theorem boundary and no external upgrade	operational	not_applicable
CM-v19-002	Infinite-bar threshold invariance	proved	not_applicable
CM-v19-003	Hard threshold deletion idempotence	proved	not_applicable
CM-v19-004	Hard threshold composition	proved	not_applicable
CM-v19-005	Retained-profile threshold monotonicity	proved	not_applicable
CM-v19-006	Positive-threshold short bars are not Serre	proved	not_applicable
CM-v19-007	Nested window restriction identity	proved	not_applicable
CM-v19-008	Fixed window-first repaired evaluation	operational	not_applicable
CM-v19-009	No unconditional crop-collapse commutation	proved	not_applicable
CM-v19-010	Exact crop-collapse commutation criterion	conditional	fixed_exact
CM-v19-011	Global non-expansiveness of hard deletion fails	proved	not_applicable
CM-v19-012	Sharp two-sided threshold-gap-safe stability	proved	fixed_metric_gap_certified
CM-v19-013	Scalarized-ledger-safe threshold classification	conditional	fixed_metric_gap_certified
CM-v19-014	Exact comparison is not a positive-radius robustness certificate	proved	fixed_exact
CM-v19-015	Runtime comparison-mode trichotomy	operational	runtime_resolved
CM-v19-016	Overlap transfer under a scalarized threshold-gap certificate	conditional	fixed_metric_gap_certified
CM-v19-017	Windowed representative idempotence	conditional	fixed_exact
CM-v19-018	Windowed representative threshold composition	conditional	fixed_exact
CM-v19-019	Canonical two-term Ext1-edge identification	proved	not_applicable
CM-v19-020	Conditional one-way repaired PH1-to-Ext1 bridge	conditional	not_applicable
CM-v19-021	Reverse Ext1-to-repaired-profile implication fails	proved	not_applicable
CM-v19-022	Actual artifact and terminal tameness are prerequisites for Type IV	operational	not_applicable
CM-v19-023	Characterization of certified Type IV-free status	proved	not_applicable
CM-v19-024	Zero diagnostics plus no interior defects imply isomorphism	conditional	not_applicable
CM-v19-025	Full-isomorphism band implies terminal-generic stable band	conditional	not_applicable
CM-v19-026	Finite-horizon product-weighted Restart bound	proved	not_applicable
CM-v19-027	Conditional countable Restart-summability principle	conditional	not_applicable
CM-v19-028	No scalar manufacture of non-scalar evidence	proved	not_applicable
CM-v19-029	Non-compensation and dependency blocker propagation	proved	not_applicable
CM-v19-030	Finite prefixes do not certify countable readiness	proved	not_applicable
CM-v19-031	Eventually constant repaired tower with apex agreement	conditional	fixed_exact

CM ID	Title	Status	Comparison policy
CM-v19-032	Representative-invariance of eligible Ext1 discharge	conditional	fixed_exact
CM-v19-033	Exact-conjugacy invariance of Type IV diagnostics	proved	fixed_exact
CM-v19-034	Terminal-generic index identity	proved	not_applicable
CM-v19-035	Direct-sum additivity of Type IV diagnostics	proved	not_applicable
CM-v19-036	Composition subadditivity of Type IV diagnostics	proved	not_applicable
CM-v19-037	Snake-lemma control over short exact comparison diagrams	conditional	not_applicable
CM-v19-038	Single-morphism characterization of one-parameter interleaving	proved	not_applicable
CM-v19-039	Finite-trivial defect implies terminal-generic cleanliness	conditional	not_applicable
CM-v19-040	Terminal-generic cleanliness does not imply quantitative interleaving	proved	not_applicable
CM-v19-041	CM register closure and source-binding discipline	operational	not_applicable

11.2 Revision and legacy migration rule

The r1 UUIDs for CM-v19-001–CM-v19-031 are superseded by their same-numbered r2 UUIDs. The r1 closure UUID CM-v19-032 is superseded by r2 CM-v19-041. Legacy CM-1–CM-21 identifiers remain available only in the versioned namespace

`ak-hdpst:18.3:cm:<legacy-id>.`

They are never resolved by bare ID. Legacy CM-12–CM-15 are rejected-and-replaced because they depended on global non-expansiveness. The machine-readable migration tables give every successor UUID.

12 Explicit exclusions

- (i) No global non-expansiveness of hard positive-threshold deletion.
- (ii) No global exactness, positive-threshold Serre localization, or strict filtered collapse functor.
- (iii) No unconditional crop–collapse or nested-window threshold commutation.
- (iv) No metric comparison without an actual pre-threshold discrepancy bound on the profiles to which deletion is applied.
- (v) No one-sided threshold-gap record as a v19 certificate.
- (vi) No exact comparison interpreted as positive-radius robustness.
- (vii) No `not_compared` mode used to discharge an invoked comparison obligation.
- (viii) No representative existence, morphism, eligibility, terminal tameness, higher coherence, Claim UUID, or artifact identity manufactured by numerical slack.
- (ix) No reverse $\text{Ext}^1 \Rightarrow \text{PH}_1$ theorem without a separate faithfulness theorem.
- (x) No Type IV number without an actual valid artifact and terminal-tame defect profiles.
- (xi) No missing Type IV data encoded as $(0, 0)$.
- (xii) No terminal-generic zero interpreted as full isomorphism without the no-interior-defect hypothesis.

- (xiii) No terminal-generic zero interpreted as a quantitative interleaving or finite defect-length bound.
- (xiv) No use of CM-v19-038 outside the pointwise finite-dimensional one-parameter regime without a separately proved extension theorem.
- (xv) No unconditional Restart constant, default $\kappa = 1$, or ordinary unweighted summability substitute.
- (xvi) No finite prefix treated as a complete countable certificate.
- (xvii) No record-level certificate gluing interpreted as construction of a global mathematical object.
- (xviii) No external-domain theorem, conjecture solution, or cryptographic security theorem.
- (xix) No normative Latin u for the Type IV cokernel diagnostic; v is canonical.
- (xx) No higher-degree PH_n -to- Ext^n bridge is asserted in this revision; it remains a separately typed priority successor candidate.

13 Primary literature

References

- [1] U. Bauer and M. Lesnick, *Induced Matchings and the Algebraic Stability of Persistence Barcodes*, Journal of Computational Geometry **6**(2) (2015), 162–191. Corollary 6.6 is the source of CM-v19-038.

14 Provenance and status

Theory	AK-HPDST v19.0.0
CM version	CM-v19.0.0-r2
Claim-register baseline	CR-v19.0.0-r2
Normative sources	Main Chapters 1–8; Foundation Appendices A–G
Imported primary source	Bauer–Lesnick 2015, Corollary 6.6, p.f.d. one-parameter regime
Machine-readable register	AK_HPDPST_v19_0_0_Appendix_CM_v19_Canonical_r2_Register .json
Canonical notation	$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}})$
Assembly policy	XeLaTeX, A4, portable TeX Live fonts, fail-fast standard-command guards

Remark 14.1 (Next dependency). Appendix TB-v19 may consume only the CM UIDs and hypotheses declared here. Eventual constancy, apex agreement, actual comparison artifacts, terminal-tameness, and the source regime of CM-v19-038 must remain explicit. No toy bridge promotes an external-domain conclusion.